NASA WEATHER DATA BASED NEURAL NETWORK GRID CONNECTED PV SYSTEM MAXIMUM POWER POINT TRACKING

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ABSTRACT

Research and development for alternative energy sources that are cleaner, renewable, and have little to no environmental impact have been pushed by the ongoing rise in energy demand, the possibility of a decline in the use of conventional petroleum fuels, and concerns about environmental degradation. Electricity from photovoltaic (PV) systems is significantly better regarded among these alternative sources as a renewable energy source such as wind power, bioenergy, tidal energy, and hydroelectric with a wide application range because it is clean, accessible, and abundant with little to no environmental effect. However, solar energy usage is significantly impacted by the landscape, location, seasonal changes, and other natural factors, such as the recurring issue of sunlight direction and irradiance changing over time. All these factors contribute to the PV system's low efficiency. Therefore, the solar panel must be operated at its maximum power point to boost the photovoltaic system's efficiency. However, the maximum power point is not stable due to the nonlinear nature of the PV cells and changes in the weather (temperature and irradiance). How do we optimize the solar panel system to work best always to produce the maximum power even when the temperature and irradiance fluctuate? The perturb and observe (P&O) algorithm is well-known and the most often used Maximum Power Point Tracking (MPPT) algorithm because of its simplicity and ease of implementation. However, in the P&O algorithm, the operational point oscillates near the maximum power point, resulting in an unending oscillation in output power. There is also a divergence from the maximum power point when weather conditions suddenly change, leading to energy loss and lower efficiency. In this research, the PV panel maximum power point was tracked with the aid of NASA weather data-based Neural Network, which is a type of computational model, and this would be based on actual weather conditions from NASA temperature and irradiance data of a specified location (Littlerock). The critical contribution of

this research is to utilize NASA weather data-based Neural Network to forecast the optimized system reference voltage in all weather conditions and to use it as a reference for the PI controller to ensure that the DC/DC boost converter generates maximum power and a steady output voltage in all weather conditions to the inverter and the grid. The effectiveness of this MPPT was compared to the Perturb & Observe (P & O) algorithms for tracking the maximum power point of solar PV panels. Using MATLAB Simulink software, the simulation results verified the proposed work's performance under diverse weather conditions.

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CHAPTER ONE

INTRODUCTION

Our everyday life depends on the steady supply of electricity. Fossil fuels were traditionally burned to produce electric power, which significantly adversely affected human health and the ecosystem on earth. As a result, lately, efforts have focused on devising alternate methods of producing electric energy using sustainable and clean energy sources, such as sunlight [1]. A solar photovoltaic (PV) cell is a device that harnesses the photovoltaic effect of semiconductors to convert energy supplied by optical and electromagnetic radiation directly to electricity [2,5]. Due to its prolonged life, low operational costs, availability, inexhaustibility, safety, and ability to be scaled to deliver the appropriate energy output, solar energy systems have become widely used as an alternative to hydrocarbon energy sources around the globe [3]. However, the PV systems' efficiency at converting energy is comparatively low (15-25%) due to some factors such as semiconductors' inherent material properties, dust, amount of solar radiation received, cell temperature, parasitic resistances, module orientation, geographical location, weather conditions, cloud, and other shading effects [4-7]. As a result of the effects of these factors, there is a need to pay attention to running the PV system at the best operating point under varied situations to lessen the PV system's power loss [7]. Maximum power point tracking (MPPT) techniques are required to extract the maximum power from the PV cell to improve PV system efficiency and produce as much electricity as possible. The algorithms continuously track the maximum power point at which maximum power prevails. There is only one maximum power point for each P-V and I-V curve at given temperature and irradiance levels [7,8]. Because the V-I characteristic of a solar cell is nonlinear and varies with temperature and irradiation, there is typically a single point known as the maximum power point on the V-I or V-P curve, at which the array, converter, and other components of the PV system operate at their highest efficiency and generate the most power. The maximum power point location is unknown, but it can be found using search algorithms or calculation models.

To keep the PV array's operating point at its maximum power point, Maximum Power Point Tracking (MPPT) methods are required. Controlling the PV systems' maximum power operation has been the subject of numerous efforts. Among the earliest approaches to addressing the issue was Incremental Conductance (IC) and Perturb and Observe (P & O). The Perturb and Observe (P & O) method is based on trial-and-error means in finding and tracing the maximum power point. The tracking controller perturbs the operating point by sweeping the operating voltage and monitoring the power variation at every cycle. From these measurements, the tracking controller derives the actual PV power [10]. The Incremental Conductance (IC) method involves differentiating the PV power with respect to voltage and the maximum power point is located when the differentiation result is zero [10]. Due to their simplicity, both approaches are widely utilized. However, they have some disadvantages of their own, such as uncertainty caused by their sensitivity to abrupt weather changes and oscillations around the operating point. To reduce the oscillations near the operating point in a steady condition, which results in the waste of some available energy, this study focus on tracking the maximum power point of a PV panel using a machine learning model which is the NASA weather data based neural network. The model would be based on actual weather conditions data from NASA temperature and irradiance data for a specific location (Littlerock).

PROBLEM STATEMENT

Due to various reasons, the efficiency of PV systems in converting energy is very low (15-25%). As a result of the effects of these factors, it is necessary to operate the PV system at the best operating point under varied situations to minimize the PV system's power loss. The perturb and observe (P&O) method has the drawbacks of not always being able to track maximum power point and having a slow response to sudden changes in solar temperature and irradiance. Because of its simplicity, Perturb and Observe (P&O) is one of the most widely used MPPT approaches; however, it has some limitations, such as uncertainty caused by its sensitivity to abrupt weather changes and oscillations around the maximum power point; thus, there is a need for a highly accurate model that not only tracks the maximum power point of a PV system but also reduces oscillations near the operating point under diverse weather conditions. The neural network is to be used because it is an artificial intelligence method that has more advantages than conventional methods such as better performance in terms of dynamic and fast response, less steady state oscillations under fast-changing weather condition.

RESEARCH AIM

This work aims to develop an intelligent maximum power point tracking (MPPT) method for a grid-connected photovoltaic (PV) system using a machine learning model which is a NASA weather data based neural network that is an artificial intelligence method that has better performance in terms of dynamic and fast response, less steady state oscillations under fast-changing weather condition. The proposed technique was implemented in MATLAB/Simulink and compared with the conventional method of Perturb and Observe (P&O).

The objectives of this work are;

1) To design a PV system that is connected to the grid, a maximum power point tracking controller, a DC-DC boost converter, and an inverter using MATLAB/Simulink software.
2) To develop a computational neural network based maximum power point tracking (MPPT) method using NASA weather data and examine how well the computational neural network-based maximum power point tracking (MPPT) method performs under changing weather conditions.

3) To compare the proposed neural network-based MPPT controller's performance with that of the perturb and observe (P&O) method and the grid-connected PV system design's performance with a PI controller.

RESEARCH OBJECTIVE

The research objectives are to create an intelligent maximum power point tracking (MPPT) method for a grid-connected photovoltaic (PV) system by employing a machine learning model that is a NASA weather data-based neural network. This artificial intelligence method has better performance in terms of dynamic and fast response, resulting in fewer steady-state oscillations in weather conditions that are rapidly changing. MATLAB/Simulink was used for the simulation. The proposed system's modeling in MATLAB/Simulink consists of seven major sections: the PV array, the DC-DC boost converter, which is the power interface section, the MPPT block, the PI controller, the Pulse width modulator (PWM), the inverter, and the grid.

The PV module's output voltage was effectively increased by means of the DC-DC boost converter, in order to carry out energy absorption and injection from the PV array to the grid-tied inverter, the boost converter served as a transmission medium. A combination of an inductor, an electronic switch, a diode, and an output capacitor carried out the energy absorption and injection process. The inverter is the component in charge of controlling the flow of electricity between the DC source and the grid. It converts the DC power from the PV array into AC power before sending it into the grid.

The objectives of this work and how they were accomplished are listed below as:

1) To design a PV system that is connected to the grid, a maximum power point tracking controller, a DC-DC boost converter, and an inverter using MATLAB/Simulink software.

- The design was created using the Simscape library tool of the Simulink software. Some blocks have been designed and can be found in the library browser of Simulink. The values for each component block were determined in accordance with the necessary calculations and specifications.

2) To develop a computational neural network based maximum power point tracking (MPPT) method using NASA weather data and examine how well the computational neural network-based maximum power point tracking (MPPT) method performs under changing weather conditions.
   - The weather (temperature and irradiance) data for the specified location (Littlerock, Arkansas) over a specified period was downloaded from the NASA website at https://power.larc.nasa.gov/data-access-viewer/. The downloaded data was used to generate input and output data for the neural network-based maximum power point tracking (MPPT) training and testing processes to build the MPPT block.
3) To compare the proposed neural network-based MPPT controller's performance with that of the perturb and observe (P&O) method and the grid-connected PV system design's performance with a PI controller.
   - The outputs of the slopes and the Simulink display were used to evaluate the performances of the MPPT blocks that have been replaced with one another.

The expected outcome of this work are: It is anticipated that the MATLAB/Simulink software simulation of the PV system with a maximum power point tracking, DC-DC converter, inverter, and grid run flawlessly and error-free. For each specified temperature and irradiance, the developed computational neural network-based maximum power point tracking (MPPT) method is expected to produce the expected maximum power of the PV array. Maximum power